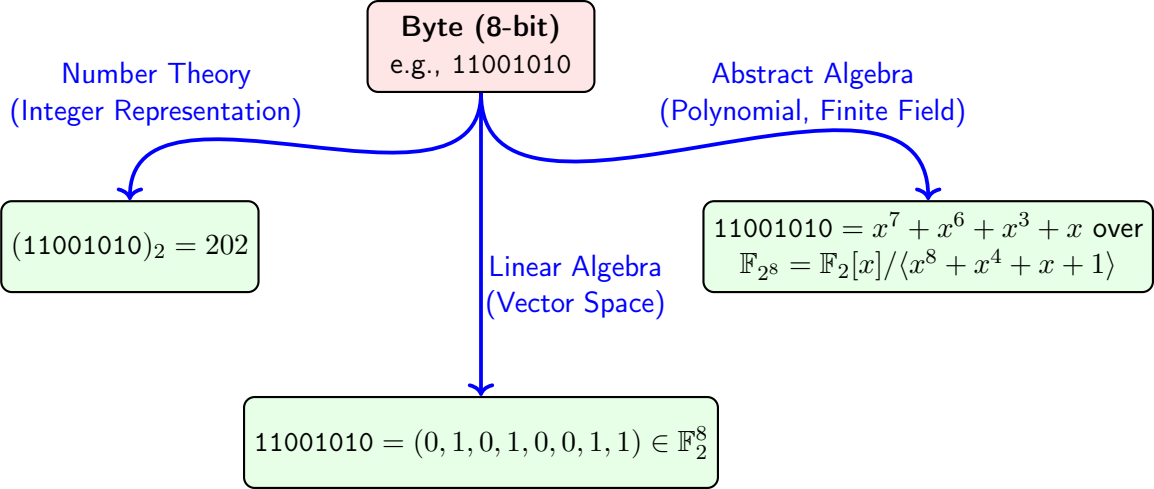




Byte (8-bit)

e.g., 11001010

Number Theory
(Integer Representation)

Abstract Algebra
(Polynomial, Finite Field)

Linear Algebra
(Vector Space)

$$(11001010)_2 = 202$$

$$11001010 = x^7 + x^6 + x^3 + x \text{ over } \mathbb{F}_{2^8} = \mathbb{F}_2[x]/\langle x^8 + x^4 + x + 1 \rangle$$

$$11001010 = (0, 1, 0, 1, 0, 0, 1, 1) \in \mathbb{F}_2^8$$